

**August 13<sup>th</sup> 2024**

## **Graphene synthesis project update**

(Update shared by Daniel Borowski as temporary stand-in for Torsten Busch during cw31 to cw33)

### **1. Biochar synthesis**

- Overall seven syntheses finished with hemp fibers from BAFA neu GmbH
  - three syntheses done with 0.02M sulfuric acid
    - first product not representative, not used further
    - second and third batch converted (see below)
  - four syntheses done with 0.10M sulfuric acid
    - very similar observations and outputs
    - products of these have been combined/homogenized
    - part of the combined lots has been converted
- Canadian Rockies Hemp arrived and cleared customs
  - several standard synthesis batches finished, seven of these (11.2 g) combined and homogenized
  - additional experiments done with higher concentrated sulfuric acid (1.5%, 2.0%, 5.0%)
  - one experiment performed under stirring, using cut up hemp fibers, product is finer and more homogeneous
  - one experiment performed with sulfuric acid pre-treatment of hemp (95 °C)
- Larger vessel reactivated, first production batch completed (14.9 g output)

### **2. Biochar analysis**

- Sample of combined biochar (milled, MB2946X\_gem) analyzed by SEM and Raman for reference

### **3. Graphene synthesis**

- two experiments finished with steep heating ramp (20-30 °C/min)
  - biochar ground manually for first experiment (TB1131)
  - laboratory mill used for second experiment (TB1132)
  - products not yet characterized
  - no serious handling issues
- two experiments performed with much gentler heating ramp (3 °C/min, TB1133/1134)
  - better handling but no other apparent difference
- one experiment performed at 5 °C/min (MB2946)

- 
- 2 g of combined biochar milled with 2 g of KOH and used for several batches on 0.25 g scale
    - 800 °C, 1 h: minimal yield (MB2947A)
    - 800 °C, 2 h: minimal yield (MB2947B)
    - 800 °C, 3 h: no output (MB2947C)
      - ➔ Inertion apparently insufficient and material likely incinerated
    - 800 °C, 3 h repeat, much higher Ar stream: 38 mg output obtained (MB2947D)
    - 800 °C, 1 h, nitrogen inertion: 60 mg output (TB1135-1)
    - 800 °C, 1 h, increased nitrogen inertion: 70 mg output (TB1135-2)
  - 2 g of combined biochar milled with 2 g of KOH and used for several batches on 0.25 g scale
    - 800 °C, 2 h, inertion equivalent to TB1135-2: 70 mg output (MB2952B)
    - 800 °C, 3 h, inertion equivalent to TB1135-2: 50 mg output (MB2952C)
      - ➔ smoldering observed upon contact of the cold material with air (see video)
    - 800 °C, 3 h, inertion equivalent to TB1135-2: 60 mg output (MB2952C2)
      - ➔ material wetted with water before/during removal from the apparatus
  - Oxygen sensor shows complete inertion of apparatus within 3 minutes at selected nitrogen stream (equivalent to TB1135-2)
  - 2 g of combined biochar milled with 3 g of KOH and used for several batches on 0.25 g scale
    - 800 °C, 1 h, standard inertion: Quartz tube shattered, only 7 mg output (MB2955A)
    - 800 °C, 1 h, standard inertion (repeat of MB2955A): 90 mg output (MB2955A2)
    - 800 °C, 2 h, standard inertion: 80 mg output (TB1136)
  - 2 g of combined biochar milled with 4 g of KOH and used for several batches on 0.25 g scale
    - 800 °C, 1 h, standard inertion: 80 mg output (MB2962A)
    - 800 °C, 2 h, standard inertion: 90 mg output (MB2962B)
  - 0.6 g of biochar (cut up, hydrolyzed under stirring) milled with 0.6 g of KOH and used for two batches on 0.2 – 0.3 g scale
    - 800 °C, 1 h, standard inertion: 70 mg output (MB2963A)
    - 800 °C, 2 h, standard inertion: 120 mg output (MB2963B)
  - 2 g of combined biochar milled with 8 g of KOH and used for several batches on 0.25 g scale
    - 800 °C, 1 h, standard inertion: 60 mg output (TB1137-1)
    - 800 °C, 2 h, standard inertion: 60 mg output (TB1137-2)
    - 800 °C, 1 h, standard inertion: 70 mg output (TB1137-3)
  - 2 g of biochar lot MB2959 milled with 12 g of KOH and used for several batches
    - 800 °C, 1 h, standard inertion: 24 mg output (MB2965A)
    - 800 °C, 2 h, standard inertion: 13 mg output (MB2965B)
    - 800 °C, 2 h, standard inertion (oven B): 7 mg output (MB2965B2)
  - 2 g of biochar lot MB2959 milled with 8 g of KOH and used for several batches
    - 800 °C, 1 h, standard inertion: 52 mg output (MB2966A)
    - 800 °C, 2 h, standard inertion (oven B): 46 mg output (MB2966B)
-

- 1.2 g of biochar lot MB2964 milled with 4.8 g of KOH and used for several batches
  - 800 °C, 1 h, standard inertion: 57 mg output (MB2967A)
  - 800 °C, 2 h, standard inertion (oven B): 42 mg output (MB2967B)
  
- 1.4 g of biochar lot MB2957 (from 1.5% H<sub>2</sub>SO<sub>4</sub>) milled with 2.1 g of KOH and used for several batches
  - 800 °C, 1 h, standard inertion: 63 mg output (MB2970A)
  - 800 °C, 2 h, standard inertion (oven B): 19 mg output (MB2970B)
  
- 1.4 g of biochar lot MB2958 (from 2% H<sub>2</sub>SO<sub>4</sub>) milled with 2.1 g of KOH and used for several batches
  - 800 °C, 1 h, standard inertion: 54 mg output (MB2971A)
  - 800 °C, 2 h, standard inertion (oven B): 66 mg output (MB2971B)
  
- 1.4 g of biochar lot MB2961 (from 5% H<sub>2</sub>SO<sub>4</sub>) milled with 2.1 g of KOH and used for several batches
  - 800 °C, 1 h, standard inertion: 88 mg output (MB2972A)
  - 800 °C, 2 h, standard inertion (oven B): 57 mg output (MB2972B)
  
- 1.4 g of biochar lot MB2959 milled with 2.1 g of KOH and used for two identical batches (reproducibility experiment)
  - 800 °C, 1 h, standard inertion: 42 mg output (MB2973A)
  - 800 °C, 1 h, standard inertion (oven B): 90 mg output (MB2973B)
  
- 1.4 g of biochar lot MB2959 milled with 2.1 g of KOH and used for two batches (milling time experiment)
  - milling time 1 min, 800 °C, 1 h, standard inertion: in progress (MB2974A)
  - milling time 5 min, 800 °C, 1 h, standard inertion (oven B): in progress (MB2974B)

#### **4. Graphene analysis**

- SEM (and partially Raman) results available for
  - TB1133, MB2946A, MB2947D, TB1135-1, TB1135-2, MB2952B, MB2952C, MB2952C2, MB2955A, MB2955A2, TB1136, MB2962A/B, MB2963A/B, TB1137-1/2/3 (step 2 products), MB2965B, MB2965B2, MB2966A/B, MB2967A/B, MB2970A/B, MB2971A/B, MB2972A/B, MB2973A/B
- 900561-500MG (Sigma-Aldrich reference)
- 924458-500MG (Sigma-Aldrich reference "Single-layer graphene sheets for battery, Bio- sourced")
  
- NDA with TEM/BET provider signed, shipment of first sample in progress

#### **5. Currently open questions**

Planned experiments:

- Use KOH/biochar pellets in step 2 (improved contact between KOH and biochar in dense pellets?)
- Step 1 material required for further testing: repeat step 1 big batch

## Experiment overview – Step 1

| Experiment | Reactor                                                                      | Raw material |                        |         | Acid  |        |               | T      | t      | pressure |             | Wash  |       | Output |                                            |  |
|------------|------------------------------------------------------------------------------|--------------|------------------------|---------|-------|--------|---------------|--------|--------|----------|-------------|-------|-------|--------|--------------------------------------------|--|
| MB2928     | AV1                                                                          | 6.0 g        | BAFA neu Hemp Fibre VF | 100.0 g | 0.19% | 0.02 M | Sulfuric acid | 180 °C | 22.5 h | 8.7 bar  | -> 10.4 bar | 742 g | Water | 2.6 g  |                                            |  |
| Comment    | Fibres not completely covered by liquid                                      |              |                        |         |       |        |               |        |        |          |             |       |       |        |                                            |  |
| MB2933     | AV1                                                                          | 6.1 g        | BAFA neu Hemp Fibre VF | 174.0 g | 0.19% | 0.02 M | Sulfuric acid | 180 °C | 24 h   | 8.7 bar  | -> 9.5 bar  | 647 g | Water | 1.2 g  |                                            |  |
| Comment    | dark crusts on glassware, metal and liquid surface removed before filtration |              |                        |         |       |        |               |        |        |          |             |       |       |        |                                            |  |
| MB2935     | AV1                                                                          | 6.0 g        | BAFA neu Hemp Fibre VF | 172.9 g | 0.19% | 0.02 M | Sulfuric acid | 180 °C | 23 h   | 8.8 bar  | -> 9.5 bar  | 500 g | Water | 1.3 g  |                                            |  |
| Comment    | dark crusts on glassware, metal and liquid surface removed before filtration |              |                        |         |       |        |               |        |        |          |             |       |       |        |                                            |  |
| MB2936     | AV1                                                                          | 6.0 g        | BAFA neu Hemp Fibre VF | 172.4 g | 0.96% | 0.10 M | Sulfuric acid | 180 °C | 23 h   | 8.8 bar  | -> 10.6 bar | 589 g | Water | 0.8 g  | Combined and homogenized as lot no MRa231  |  |
| Comment    | dark crusts on glassware, metal and liquid surface removed before filtration |              |                        |         |       |        |               |        |        |          |             |       |       |        |                                            |  |
| MRa225     | AV1                                                                          | 6.0 g        | BAFA neu Hemp Fibre VF | 153.8 g | 0.96% | 0.10 M | Sulfuric acid | 180 °C | 24 h   | ?        | -> 10.5 bar | 500 g | Water | 0.9 g  |                                            |  |
| Comment    | dark crusts on glassware, metal and liquid surface removed before filtration |              |                        |         |       |        |               |        |        |          |             |       |       |        |                                            |  |
| MRa228     | AV1                                                                          | 6.3 g        | BAFA neu Hemp Fibre VF | 154.0 g | 0.96% | 0.10 M | Sulfuric acid | 180 °C | 22.5 h | 8.7 bar  | -> 11.0 bar | 500 g | Water | 0.9 g  |                                            |  |
| Comment    | dark crusts on glassware, metal and liquid surface removed before filtration |              |                        |         |       |        |               |        |        |          |             |       |       |        |                                            |  |
| MRa231     | AV1                                                                          | 6.1 g        | BAFA neu Hemp Fibre VF | 168.8 g | 0.96% | 0.10 M | Sulfuric acid | 180 °C | 23.5 h | 8.0 bar  | -> 10.5 bar | 500 g | Water | 0.8 g  |                                            |  |
| Comment    | dark crusts on glassware, metal and liquid surface removed before filtration |              |                        |         |       |        |               |        |        |          |             |       |       |        |                                            |  |
| MB2937     | AV1                                                                          | 9.4 g        | Canadian Rockies Hemp  | 179.3 g | 0.96% | 0.10 M | Sulfuric acid | 180 °C | 24 h   | 8.6 bar  | -> 10.9 bar | 569 g | Water | 1.6 g  | Combined and homogenized as lot no MB2946X |  |
| Comment    | very low amount of foreign material observed and removed during filtration   |              |                        |         |       |        |               |        |        |          |             |       |       |        |                                            |  |
| MB2938     | AV1                                                                          | 8.9 g        | Canadian Rockies Hemp  | 175.0 g | 0.96% | 0.10 M | Sulfuric acid | 180 °C | 23.5 h | 9.2 bar  | -> 13.2 bar | 577 g | Water | 1.4 g  |                                            |  |
| Comment    | very low amount of foreign material observed and removed during filtration   |              |                        |         |       |        |               |        |        |          |             |       |       |        |                                            |  |
| MB2940     | AV1                                                                          | 9.2 g        | Canadian Rockies Hemp  | 177.1 g | 0.96% | 0.10 M | Sulfuric acid | 180 °C | 22.5 h | 8.7 bar  | -> 10.6 bar | 555 g | Water | 1.5 g  |                                            |  |
| Comment    | very low amount of foreign material observed and removed during filtration   |              |                        |         |       |        |               |        |        |          |             |       |       |        |                                            |  |
| MB2942     | AV1                                                                          | 8.9 g        | Canadian Rockies Hemp  | 177.6 g | 0.96% | 0.10 M | Sulfuric acid | 180 °C | 23 h   | 8.3 bar  | -> 13.5 bar | 578 g | Water | 1.6 g  |                                            |  |
| Comment    | low amount of foreign material observed and removed during filtration        |              |                        |         |       |        |               |        |        |          |             |       |       |        |                                            |  |
| MB2943     | AV1                                                                          | 9.8 g        | Canadian Rockies Hemp  | 178.2 g | 0.96% | 0.10 M | Sulfuric acid | 180 °C | 24 h   | 8.3 bar  | -> 12.7 bar | 611 g | Water | 1.7 g  |                                            |  |
| Comment    | low amount of foreign material observed and removed during filtration        |              |                        |         |       |        |               |        |        |          |             |       |       |        |                                            |  |
| MB2944     | AV1                                                                          | 9.0 g        | Canadian Rockies Hemp  | 178.4 g | 0.96% | 0.10 M | Sulfuric acid | 180 °C | 24 h   | 8.4 bar  | -> 9.7 bar  | 593 g | Water | 1.8 g  |                                            |  |
| Comment    | low amount of foreign material observed and removed during filtration        |              |                        |         |       |        |               |        |        |          |             |       |       |        |                                            |  |
| MB2945     | AV1                                                                          | 9.1 g        | Canadian Rockies Hemp  | 177.7 g | 0.96% | 0.10 M | Sulfuric acid | 180 °C | 24 h   | 8.7 bar  | -> 9.8 bar  | 6 g   | Water | 1.6 g  |                                            |  |
| Comment    | low amount of foreign material observed and removed during filtration        |              |                        |         |       |        |               |        |        |          |             |       |       |        |                                            |  |
| MB2948     | AV1                                                                          | 9.4 g        | Canadian Rockies Hemp  | 175.1 g | 0.96% | 0.10 M | Sulfuric acid | 180 °C | 25 h   | 9.0 bar  | -> 12.6 bar | 475 g | Water | 1.7 g  |                                            |  |
| Comment    | low amount of foreign material observed and removed during filtration        |              |                        |         |       |        |               |        |        |          |             |       |       |        |                                            |  |

| Experiment | Reactor                                                                                           | Raw material                      | Acid                                         | T      | t      | pressure             | Wash                | Output |
|------------|---------------------------------------------------------------------------------------------------|-----------------------------------|----------------------------------------------|--------|--------|----------------------|---------------------|--------|
| MB2949     | AV1                                                                                               | 9.2 g Canadian Rockies Hemp       | 174.3 g 0.96% 0.10 M Sulfuric acid           | 180 °C | 23.5 h | 8.4 bar -> 10.2 bar  | 538 g Water         | 1.8 g  |
| Comment    | low amount of foreign material observed and removed during filtration                             |                                   |                                              |        |        |                      |                     |        |
| MB2950     | AV1                                                                                               | 9.1 g Canadian Rockies Hemp       | 170.5 g 0.96% 0.10 M Sulfuric acid           | 180 °C | 23 h   | -> 11.1 bar          | 534 g Water         | 1.6 g  |
| Comment    | low amount of foreign material observed and removed during filtration                             |                                   |                                              |        |        |                      |                     |        |
| MB2951     | AV1                                                                                               | 9.3 g Canadian Rockies Hemp       | 178.5 g 0.96% 0.10 M Sulfuric acid           | 180 °C | 24 h   | 9.1 bar -> 11.2 bar  | 575 g Water         | 1.8 g  |
| Comment    | low amount of foreign material observed and removed during filtration                             |                                   |                                              |        |        |                      |                     |        |
| MRa255     | AV1                                                                                               | 9.2 g Canadian Rockies Hemp       | 170.8 g 0.96% 0.10 M Sulfuric acid           | 180 °C | 23.5 h | 8.1 bar -> 12.8 bar  | 488 g Water         | 1.7 g  |
| Comment    | some foreign material observed and removed during filtration                                      |                                   |                                              |        |        |                      |                     |        |
| MB2953     | AV1                                                                                               | 9.7 g Canadian Rockies Hemp       | 177.1 g 0.96% 0.10 M Sulfuric acid           | 180 °C | 23.5 h | 9.1 bar -> 10.4 bar  | 558 g Water         | 1.8 g  |
| Comment    | low amount of foreign material observed and removed during filtration                             |                                   |                                              |        |        |                      |                     |        |
| MB2956     | AV1                                                                                               | 9.8 g Canadian Rockies Hemp       | 172.0 g 0.96% 0.10 M Sulfuric acid           | 180 °C | 24 h   | 8.6 bar -> 12.5 bar  | 548 g Water         | 1.9 g  |
| Comment    | low amount of foreign material observed and removed during filtration                             |                                   |                                              |        |        |                      |                     |        |
| MB2957     | AV1                                                                                               | 9.1 g Canadian Rockies Hemp       | 176.8 g 1.50% 0.15 M Sulfuric acid           | 180 °C | 23.5 h | 8.8 bar -> 8.5 bar   | 589 g Water         | 1.5 g  |
| Comment    |                                                                                                   |                                   |                                              |        |        |                      |                     |        |
| MB2958     | AV1                                                                                               | 9.3 g Canadian Rockies Hemp       | 177.0 g 2.00% 0.20 M Sulfuric acid           | 180 °C | 24 h   | 8.9 bar -> 8.7 bar   | 562 g Water         | 1.8 g  |
| Comment    |                                                                                                   |                                   |                                              |        |        |                      |                     |        |
| MB2959     | AV5                                                                                               | 76.0 g Canadian Rockies Hemp      | 953.0 g 1.00% 0.10 M Sulfuric acid           | 180 °C | 25 h   |                      | 3957 g Water        | 14.9 g |
| Comment    | low amount of foreign material observed and removed during filtration                             |                                   |                                              |        |        |                      |                     |        |
| MB2960     | AV1                                                                                               | 3.8 g Canadian Rockies Hemp (cut) | 151.0 g 1.00% 0.10 M Sulfuric acid           | 180 °C | 24 h   | 8.8 bar -> 10.5 bar  | 300 g Water         | 0.6 g  |
| Comment    | Reaction performed with stirring                                                                  |                                   |                                              |        |        |                      |                     |        |
| MB2961     | AV1                                                                                               | 9.9 g Canadian Rockies Hemp       | 172.0 g 5.00% 0.51 M Sulfuric acid           | 180 °C | 24 h   | 10.1 bar -> 16.1 bar | 563 g Water         | 1.7 g  |
| Comment    | low amount of foreign material observed and removed during filtration                             |                                   |                                              |        |        |                      |                     |        |
| MB2964     | AV1                                                                                               | 9.0 g Canadian Rockies Hemp       | 211.0 g + 140.8 g 1.00% 0.10 M Sulfuric acid | 180 °C | 24 h   | 8.6 bar -> 8.8 bar   | 310 g + 603 g Water | 1.3 g  |
| Comment    | Hemp boiled with 1% H2SO4 at 95 °C, filtered and washed, then heated in Autoclave with fresh acid |                                   |                                              |        |        |                      |                     |        |

## Experiment overview – Step 2

| Experiment | Oven | Qty                                                       | Lot     | Base             | grinding     | Gas | T (rate)     | T (max) | t   | Wash                   | Drying                           | Output   |
|------------|------|-----------------------------------------------------------|---------|------------------|--------------|-----|--------------|---------|-----|------------------------|----------------------------------|----------|
| TB1131     | A    | 1.2 g                                                     | MB2933  | 1.2 g KOH (90%)  | manual       | Ar  | 20-27 °C/min | 790 °C  | 1 h | 20 g 10% HCl (+ water) | 100 °C N2 stream (atm. Pressure) | 0.16 g   |
| Comment    |      | ground biochar: brown powder, strong electrostatic charge |         |                  |              |     |              |         |     |                        |                                  |          |
| TB1132     | A    | 1.3 g                                                     | MB2935  | 1.3 g KOH (90%)  | mill (2 min) | Ar  | 20-29 °C/min | 792 °C  | 1 h | 20 g 10% HCl (+ water) | 100 °C N2 stream (atm. Pressure) | 0.3 g    |
| Comment    |      | ground biochar: brown powder                              |         |                  |              |     |              |         |     |                        |                                  |          |
| TB1133     | A    | 1.0 g                                                     | MRa231  | 1.0 g KOH (90%)  | mill (1 min) | Ar  | 3 °C/min     | 785 °C  | 1 h | 20 g 10% HCl (+ water) | 100 °C N2 stream (atm. Pressure) | 0.3 g    |
| Comment    |      | ground biochar: brown powder; see temperature trend       |         |                  |              |     |              |         |     |                        |                                  |          |
| TB1134     | A    | 1.0 g                                                     | MRa231  | 1.0 g KOH (90%)  | mill (2 min) | Ar  | 3 °C/min     | 771 °C  | 1 h | 20 g 10% HCl (+ water) | 100 °C N2 stream (atm. Pressure) | 0.3 g    |
| Comment    |      | ground biochar: brown powder; see temperature trend       |         |                  |              |     |              |         |     |                        |                                  |          |
| MB2946     | A    | 1.0 g                                                     | MB2946X | 1.0 g KOH (90%)  | mill (1 min) | Ar  | 5 °C/min     | 770 °C  | 1 h | 22 g 10% HCl (+ water) | 100 °C N2 stream (atm. Pressure) | 0.3 g    |
| Comment    |      | ground biochar: brown powder; see temperature trend       |         |                  |              |     |              |         |     |                        |                                  |          |
| MB2947A    | A    | 0.25 g                                                    | MB2946X | 0.25 g KOH (90%) | mill (1 min) | Ar  | 3 °C/min     | 803 °C  | 1 h | 10 g 10% HCl (+ water) | 40 °C N2 stream 210 mbar         | < 0.01 g |
| Comment    |      | ground biochar: brown powder; see temperature trend       |         |                  |              |     |              |         |     |                        |                                  |          |
| MB2947B    | A    | 0.25 g                                                    | MB2946X | 0.25 g KOH (90%) | mill (1 min) | Ar  | 3 °C/min     | 804 °C  | 2 h | 10 g 10% HCl (+ water) | 40 °C N2 stream 210 mbar         | < 0.01 g |
| Comment    |      | ground biochar: brown powder; see temperature trend       |         |                  |              |     |              |         |     |                        |                                  |          |
| MB2947C    | A    | 0.25 g                                                    | MB2946X | 0.25 g KOH (90%) | mill (1 min) | Ar  | 3 °C/min     | 806 °C  | 3 h | Batch discarded        |                                  |          |
| Comment    |      | ground biochar: brown powder; see temperature trend       |         |                  |              |     |              |         |     |                        |                                  |          |
| MB2947D    | A    | 0.25 g                                                    | MB2946X | 0.25 g KOH (90%) | mill (1 min) | Ar  | 3 °C/min     | 810 °C  | 3 h | 10 g 10% HCl (+ water) | 100 °C N2 stream (atm. Pressure) | 38 mg    |
| Comment    |      | ground biochar: brown powder; see temperature trend       |         |                  |              |     |              |         |     |                        |                                  |          |
| TB1135-1   | A    | 0.25 g                                                    | MB2946X | 0.25 g KOH (90%) | mill (1 min) | N2  | 3 °C/min     | 800 °C  | 1 h | 14 g 10% HCl (+ water) | 100 °C N2 stream (atm. Pressure) | 0.06 g   |
| Comment    |      | ground biochar: brown powder; see temperature trend       |         |                  |              |     |              |         |     |                        |                                  |          |
| TB1135-2   | A    | 0.25 g                                                    | MB2946X | 0.25 g KOH (90%) | mill (1 min) | N2  | 3 °C/min     | 800 °C  | 1 h | 14 g 10% HCl (+ water) | 100 °C N2 stream (atm. Pressure) | 0.07 g   |
| Comment    |      | ground biochar: brown powder; see temperature trend       |         |                  |              |     |              |         |     |                        |                                  |          |
| MB2952B    | A    | 0.25 g                                                    | MB2946X | 0.25 g KOH (90%) | mill (1 min) | N2  | 3 °C/min     | 800 °C  | 2 h | 10 g 10% HCl (+ water) | 100 °C N2 stream (atm. Pressure) | 0.07 g   |
| Comment    |      | ground biochar: brown powder; see temperature trend       |         |                  |              |     |              |         |     |                        |                                  |          |
| MB2952C    | A    | 0.25 g                                                    | MB2946X | 0.25 g KOH (90%) | mill (1 min) | N2  | 3 °C/min     | 800 °C  | 3 h | 10 g 10% HCl (+ water) | 100 °C N2 stream (atm. Pressure) | 0.05 g   |
| Comment    |      | ground biochar: brown powder; see temperature trend       |         |                  |              |     |              |         |     |                        |                                  |          |

| Experiment | Oven | Qty                                                                             | Lot     | Base             | grinding       | Gas | T (rate) | T (max) | t   | Wash         | Drying           | Output |
|------------|------|---------------------------------------------------------------------------------|---------|------------------|----------------|-----|----------|---------|-----|--------------|------------------|--------|
| MB2952C2   | A    | 0.25 g                                                                          | MB2946X | 0.25 g KOH (90%) | mill (1 min)   | N2  | 3 °C/min | 800 °C  | 2 h | 10 g 10% HCl | 100 °C N2 stream | 0.06 g |
| Comment    |      | ground biochar: brown powder; see temperature trend                             |         |                  |                |     |          |         |     | (+ water)    | (atm. Pressure)  |        |
| MB2955A    | A    | 0.25 g                                                                          | MB2946X | 0.38 g KOH (90%) | mill (1 min)   | N2  | 3 °C/min | 800 °C  | 1 h | 10 g 10% HCl | 100 °C N2 stream | 7 mg   |
| Comment    |      | Quartz tube shattered during experiment                                         |         |                  |                |     |          |         |     | (+ water)    | (atm. Pressure)  |        |
| MB2955A2   | A    | 0.25 g                                                                          | MB2946X | 0.38 g KOH (90%) | mill (1 min)   | N2  | 3 °C/min | 800 °C  | 1 h | 10 g 10% HCl | 100 °C N2 stream | 0.09 g |
| Comment    |      | ground biochar: brown powder; see temperature trend                             |         |                  |                |     |          |         |     | (+ water)    | (atm. Pressure)  |        |
| TB1136     | A    | 0.26 g                                                                          | MB2946X | 0.39 g KOH (90%) | mill (1 min)   | N2  | 3 °C/min | 800 °C  | 2 h | 14 g 10% HCl | 100 °C N2 stream | 0.08 g |
| Comment    |      | ground biochar: brown powder; see temperature trend                             |         |                  |                |     |          |         |     | (+ water)    | (atm. Pressure)  |        |
| MB2962A    | A    | 0.25 g                                                                          | MB2946X | 0.50 g KOH (90%) | mill (1 min)   | N2  | 3 °C/min | 800 °C  | 1 h | 10 g 10% HCl | 100 °C N2 stream | 0.08 g |
| Comment    |      | ground biochar: brown powder; see temperature trend                             |         |                  |                |     |          |         |     | (+ water)    | (atm. Pressure)  |        |
| MB2962B    | A    | 0.25 g                                                                          | MB2946X | 0.50 g KOH (90%) | mill (1 min)   | N2  | 3 °C/min | 800 °C  | 2 h | 10 g 10% HCl | 100 °C N2 stream | 0.09 g |
| Comment    |      | ground biochar: brown powder; see temperature trend                             |         |                  |                |     |          |         |     | (+ water)    | (atm. Pressure)  |        |
| MB2963A    | A    | 0.22 g                                                                          | MB2960  | 0.22 g KOH (90%) | mill (1 min)   | N2  | 3 °C/min | 800 °C  | 1 h | 10 g 10% HCl | 100 °C N2 stream | 0.07 g |
| Comment    |      | ground biochar: brown powder; see temperature trend                             |         |                  |                |     |          |         |     | (+ water)    | (atm. Pressure)  |        |
| MB2963B    | A    | 0.29 g                                                                          | MB2960  | 0.29 g KOH (90%) | mill (1 min)   | N2  | 3 °C/min | 800 °C  | 2 h | 10 g 10% HCl | 100 °C N2 stream | 0.12 g |
| Comment    |      | ground biochar: brown powder; see temperature trend                             |         |                  |                |     |          |         |     | (+ water)    | (atm. Pressure)  |        |
| TB1137-1   | A    | 0.26 g                                                                          | MB2946X | 1.06 g KOH (90%) | mill (1 min)   | N2  | 3 °C/min | 800 °C  | 1 h | 23 g 10% HCl | 100 °C N2 stream | 0.06 g |
| Comment    |      | ground biochar: brown powder; see temperature trend                             |         |                  |                |     |          |         |     | (+ water)    | (atm. Pressure)  |        |
| TB1137-2   | A    | 0.28 g                                                                          | MB2946X | 1.13 g KOH (90%) | mill (1 min)   | N2  | 3 °C/min | 800 °C  | 2 h | 23 g 10% HCl | 100 °C N2 stream | 0.06 g |
| Comment    |      | ground biochar: brown powder; see temperature trend                             |         |                  |                |     |          |         |     | (+ water)    | (atm. Pressure)  |        |
| TB1137-3   | A    | 0.26 g                                                                          | MB2946X | 1.06 g KOH (90%) | mill (1 min)   | N2  | 3 °C/min | 800 °C  | 1 h | 23 g 10% HCl | 100 °C N2 stream | 0.07 g |
| Comment    |      | ground biochar: brown powder; see temperature trend                             |         |                  |                |     |          |         |     | (+ water)    | (atm. Pressure)  |        |
| MB2965A    | A    | 0.22 g                                                                          | MB2959  | 1.08 g KOH (90%) | mill (2.5 min) | N2  | 3 °C/min | 800 °C  | 1 h | 20 g 10% HCl | 100 °C N2 stream | 0.02 g |
| Comment    |      | ground biochar/KOH mixture likely absorbed some moisture; see temperature trend |         |                  |                |     |          |         |     | (+ water)    | (atm. Pressure)  |        |
| MB2965B    | A    | 0.22 g                                                                          | MB2959  | 1.08 g KOH (90%) | mill (2.5 min) | N2  | 3 °C/min | 800 °C  | 2 h | 20 g 10% HCl | 100 °C N2 stream | 0.01 g |
| Comment    |      | ground biochar/KOH mixture likely absorbed some moisture; see temperature trend |         |                  |                |     |          |         |     | (+ water)    | (atm. Pressure)  |        |
| MB2965B2   | B    | 0.22 g                                                                          | MB2959  | 1.08 g KOH (90%) | mill (2.5 min) | N2  | 3 °C/min | 800 °C  | 2 h | 20 g 10% HCl | 100 °C N2 stream | 0.01 g |
| Comment    |      | ground biochar/KOH mixture likely absorbed some moisture; see temperature trend |         |                  |                |     |          |         |     | (+ water)    | (atm. Pressure)  |        |

|         |   |                                                     |        |        |           |                |    |          |        |     |           |         |                 |           |        |
|---------|---|-----------------------------------------------------|--------|--------|-----------|----------------|----|----------|--------|-----|-----------|---------|-----------------|-----------|--------|
| MB2966A | A | 0.26 g                                              | MB2959 | 1.04 g | KOH (90%) | mill (2.5 min) | N2 | 3 °C/min | 800 °C | 1 h | 20 g      | 10% HCl | 100 °C          | N2 stream | 0.05 g |
| Comment |   | ground biochar: brown powder; see temperature trend |        |        |           |                |    |          |        |     | (+ water) |         | (atm. Pressure) |           |        |
| MB2966B | B | 0.26 g                                              | MB2959 | 1.04 g | KOH (90%) | mill (2.5 min) | N2 | 3 °C/min | 800 °C | 2 h | 20 g      | 10% HCl | 100 °C          | N2 stream | 0.05 g |
| Comment |   | ground biochar: brown powder; see temperature trend |        |        |           |                |    |          |        |     | (+ water) |         | (atm. Pressure) |           |        |
| MB2967A | A | 0.26 g                                              | MB2964 | 1.04 g | KOH (90%) | mill (2.5 min) | N2 | 3 °C/min | 800 °C | 1 h | 20 g      | 10% HCl | 100 °C          | N2 stream | 0.06 g |
| Comment |   | ground biochar: brown powder; see temperature trend |        |        |           |                |    |          |        |     | (+ water) |         | (atm. Pressure) |           |        |
| MB2967B | B | 0.26 g                                              | MB2964 | 1.04 g | KOH (90%) | mill (2.5 min) | N2 | 3 °C/min | 800 °C | 2 h | 20 g      | 10% HCl | 100 °C          | N2 stream | 0.04 g |
| Comment |   | ground biochar: brown powder; see temperature trend |        |        |           |                |    |          |        |     | (+ water) |         | (atm. Pressure) |           |        |
| MB2970A | A | 0.24 g                                              | MB2957 | 0.36 g | KOH (90%) | mill (2.5 min) | N2 | 3 °C/min | 800 °C | 1 h | 10 g      | 10% HCl | 100 °C          | N2 stream | 0.06 g |
| Comment |   | ground biochar: brown powder; see temperature trend |        |        |           |                |    |          |        |     | (+ water) |         | (atm. Pressure) |           |        |
| MB2970B | B | 0.24 g                                              | MB2957 | 0.36 g | KOH (90%) | mill (2.5 min) | N2 | 3 °C/min | 800 °C | 2 h | 10 g      | 10% HCl | 100 °C          | N2 stream | 0.02 g |
| Comment |   | ground biochar: brown powder; see temperature trend |        |        |           |                |    |          |        |     | (+ water) |         | (atm. Pressure) |           |        |
| MB2971A | A | 0.24 g                                              | MB2958 | 0.36 g | KOH (90%) | mill (2.5 min) | N2 | 3 °C/min | 800 °C | 1 h | 10 g      | 10% HCl | 100 °C          | N2 stream | 0.05 g |
| Comment |   | ground biochar: brown powder; see temperature trend |        |        |           |                |    |          |        |     | (+ water) |         | (atm. Pressure) |           |        |
| MB2971B | B | 0.24 g                                              | MB2958 | 0.36 g | KOH (90%) | mill (2.5 min) | N2 | 3 °C/min | 800 °C | 2 h | 10 g      | 10% HCl | 100 °C          | N2 stream | 0.07 g |
| Comment |   | ground biochar: brown powder; see temperature trend |        |        |           |                |    |          |        |     | (+ water) |         | (atm. Pressure) |           |        |
| MB2972A | A | 0.24 g                                              | MB2961 | 0.36 g | KOH (90%) | mill (2.5 min) | N2 | 3 °C/min | 800 °C | 1 h | 10 g      | 10% HCl | 100 °C          | N2 stream | 0.09 g |
| Comment |   | ground biochar: brown powder; see temperature trend |        |        |           |                |    |          |        |     | (+ water) |         | (atm. Pressure) |           |        |
| MB2972B | B | 0.24 g                                              | MB2961 | 0.36 g | KOH (90%) | mill (2.5 min) | N2 | 3 °C/min | 800 °C | 2 h | 10 g      | 10% HCl | 100 °C          | N2 stream | 0.06 g |
| Comment |   | ground biochar: brown powder; see temperature trend |        |        |           |                |    |          |        |     | (+ water) |         | (atm. Pressure) |           |        |
| MB2973A | A | 0.24 g                                              | MB2959 | 0.36 g | KOH (90%) | mill (2.5 min) | N2 | 3 °C/min | 800 °C | 1 h | 10 g      | 10% HCl | 100 °C          | N2 stream | 0.04 g |
| Comment |   | ground biochar: brown powder; see temperature trend |        |        |           |                |    |          |        |     | (+ water) |         | (atm. Pressure) |           |        |
| MB2973B | B | 0.24 g                                              | MB2959 | 0.36 g | KOH (90%) | mill (2.5 min) | N2 | 3 °C/min | 800 °C | 1 h | 10 g      | 10% HCl | 100 °C          | N2 stream | 0.09 g |
| Comment |   | ground biochar: brown powder; see temperature trend |        |        |           |                |    |          |        |     | (+ water) |         | (atm. Pressure) |           |        |